CUNEF UNIVERSIDAD

Econometrics 2024/2025

Topic 2: Multiple Regression Analysis: Estimation

Outline

Multiple Regression Analysis: Motivation

OLS estimation in Multiple Regression

Properties of OLS in multiple regression

Goodness-of-Fit

Assumptions for the multiple regression model

Sampling variances of the OLS estimators

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Multiple Regression Analysis

- In a multiple regression analysis we are interested in "explaining y in terms of variables x_1, x_2, x_3,...,x_k",

Slope parameters

Intercept

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + u$$

Error term

Dependent variable, explained variable, response variable, ...

Independent variables, explanatory variables, regressors, ...

Motivation for multiple regression

- Incorporate more explanatory factors into the model
- Explicitly hold other factors fixed while examining the effects of a particular independent variable on the dependent variable.
- Allow for more flexible functional forms
- **Example: Wage equation**

  β_1 now measures effect of education <u>explicitly holding experience fixed</u>

$$wage = \beta_0 + \beta_1 educ + \beta_2 exper + u$$

All other factors

Hourly wage

Years of labor market experience

Motivation for multiple regression

- Example: CEO salary, sales and CEO tenure

$$\log(salary) = \beta_0 + \beta_1 \log(sales) + \beta_2 ceoten + \beta_3 ceoten^2 + u$$

Log of CEO salary — Log sales — Quadratic function of CEO tenure with the firm

- Model assumes a quadratic relationship between CEO salary and his or her tenure with the firm

$$\frac{\partial \log(\text{salary})}{\partial \text{ceoten}} = \beta_2 + 2\beta_3 \text{ceoten}$$

  The increase in salary due to additional years of tenure depends on how many years the CEO has already been with the firm.

- The model has to be linear in the parameters (not in the variables!).

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OLS estimation in Multiple Regression Analysis

- Random sample

$$\{(x_{i1}, x_{i2}, \ldots, x_{ik}, y_i): \quad i = 1, \ldots, n\}$$

- Regression residuals

$$\widehat{u}_i = y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_{i1} - \widehat{\beta}_2 x_{i2} - \ldots - \widehat{\beta}_k x_{ik}$$

- Minimize sum of squared residuals

$$\min \text{SSR} = \min \sum_{i=1}^{n} \widehat{u}_i^2 \rightarrow \widehat{\beta}_0, \widehat{\beta}_1, \widehat{\beta}_2, \ldots, \widehat{\beta}_k$$

Interpretation of the multiple regression model

$$\beta_j = \frac{\Delta y}{\Delta x_j} \longleftarrow$$

By how much does the dependent variable change if the j-th independent variable is increased by one unit, holding all other independent variables constant

- The multiple linear regression model manages to hold the values of other explanatory variables fixed even if, in reality, they are correlated with the explanatory variable under consideration.
- "Ceteris paribus" interpretation
- It has still to be assumed that unobserved factors do not change if the explanatory variables are changed.

Interpretation of the multiple regression model

- Example: Determinants of college GPA

$$\widehat{colGPA} = 1.29 + .453\ hsGPA + .0094\ ACT$$

Grade point average at college — High school grade point average — Achievement test score

- Interpretation
  - Holding ACT fixed, another point on high school grade point average is associated with another .453 points college grade point average
  - Or: If we compare two students with the same ACT, but the hsGPA of student A is one point higher, we predict student A to have a colGPA that is .453 higher than that of student B
  - Holding high school grade point average fixed, another 10 points on ACT are associated with less than one point on college GPA

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Properties of OLS on any sample of data

1. Fitted (predicted) values

$$\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_{i1} + \widehat{\beta}_2 x_{i2} + \ldots + \widehat{\beta}_k x_{ik}$$

2. Residuals (deviations from the regression line)

$$\widehat{u}_i = y_i - \widehat{y}_i$$

3. Deviations from regression line sum up to zero:

$$\sum_{i=1}^{n} \widehat{u}_i = 0$$

4. Covariance between deviations and regressors is zero

$$\sum_{i=1}^{n} x_i \widehat{u}_i = 0$$

5. Sample averages of y and the regressors lie on regression line

$$\bar{y} = \widehat{\beta}_0 + \widehat{\beta}_1 \bar{x}_1 + \ldots + \widehat{\beta}_k \bar{x}_k$$

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Goodness-of-Fit

- Decomposition of total variation

$$SST = SSE + SSR$$

- R-squared: $R^2 \equiv SSE/SST = 1 - SSR/SST$
- Notice that R-squared can only increase if another explanatory variable is added to the regression
- Alternative expression for R-squared

$$R^2 = \frac{\left(\sum_{i=1}^{n}(y_i - \bar{y})\left(\widehat{y}_i - \overline{\widehat{y}}\right)\right)^2}{\left(\sum_{i=1}^{n}(y_i - \bar{y})^2\right)\left(\sum_{i=1}^{n}\left(\widehat{y}_i - \overline{\widehat{y}}\right)^2\right)}$$

R-squared is equal to the squared correlation coefficient between the actual and the predicted value of the dependent variable

Goodness-of-Fit

- The following regression model aims to explain the ROE in terms of several financial metrics:

$$\begin{aligned}\widehat{\text{ROE}} =& 0.153 - 0.075 \times \text{Leverage} + 0.032 \times \text{Asset Turnover} \\ & - 0.014 \times \text{Operating Expenses} + 0.062 \times \text{Market Share} \\ & R^2 = 0.0413\end{aligned}$$

- Interpretation:
  - If leverage increases by 1 unit, the predicted ROE decreases by 7.5 percentage points.
  - If asset turnover increases by 1 unit, the predicted ROE increases by 3.2 percentage points.
  - Higher operating expenses decrease ROE, while greater market share improves ROE.

Goodness-of-Fit

- An additional explanatory variable is added: Dividend Payout Ratio

$$\begin{aligned}\widehat{\text{ROE}} =& 0.153 - 0.075 \times \text{Leverage} + 0.032 \times \text{Asset Turnover} \\ & - 0.014 \times \text{Operating Expenses} + 0.062 \times \text{Market Share} \\ & + 0.025 \times \text{Dividend Payout Ratio} \\ & R^2 = 0.0422 \end{aligned} \tag{1}$$

- Interpretation:
  - The dividend payout ratio has a small positive effect on ROE.
  - The additional explanatory power of the model is limited, as R-squared increases only slightly.

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Standard assumptions for the multiple regression model

- **Assumption MLR.1** (Linear in parameters): In the population, the relationship between y and the explanatory variables is linear

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k + u$$

- **Assumption MLR.2** (Random sampling): The data is a random sample drawn from the population

$$\{(x_{i1}, x_{i2}, \ldots, x_{ik}, y_i) : i = 1, \ldots, n\}$$

Each data point therefore follows the population equation

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \ldots + \beta_k x_{ik} + u_i$$

Standard assumptions for the multiple regression model

- **Assumption MLR.3** (No perfect collinearity): In the sample (and therefore in the population), none of the independent variables is constant and there are no exact linear relationships among the independent variables.
- Remarks on MLR.3
  - The assumption only rules out perfect collinearity/correlation between explanatory variables; imperfect correlation is allowed.
  - If an explanatory variable is a perfect linear combination of other explanatory variables it is superfluous and may be eliminated.
  - Constant variables are also ruled out (collinear with intercept).

Standard assumptions for the multiple regression model

- Example for perfect collinearity: small sample

$$\text{avgscore} = \beta_0 + \beta_1 \text{ expend} + \beta_2 \text{ avginc} + u$$

  In a small sample, avginc may accidentally be an exact multiple of expend; it will not be possible to disentangle their separate effects because there is exact covariation

- Example for perfect collinearity: relationships between regressors

$$\text{vote } A = \beta_0 + \beta_1 \text{ share } A + \beta_2 \text{ share } B + u$$

  Either shareA or shareB will have to be dropped from the regression because there is an exact linear relationship between them: share A + share $B = 1$

Standard assumptions for the multiple regression model

- **Assumption MLR.4** (Zero conditional mean): The value of the explanatory variables must contain no information about the mean of the unobserved factors

$$E\left(u_i \mid x_{i1}, x_{i2}, \ldots, x_{ik}\right) = 0$$

In a multiple regression model, the zero conditional mean assumption is much more likely to hold because fewer things end up in the error.

Including irrelevant variables in a regression model

- Consider the multiple regression model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$$

  Suppose that, in the population, variable x_3 is irrelevant such that $\beta_3 = 0$.
- Since OLS estimates are unbiased,

$$E\left(\widehat{\beta}_3\right) = \beta_3 = 0$$

  There is no problem in including irrelevant variables.
- Warning: However, including irrevelant variables may increase sampling variance.

Omitting relevant variables: the simple case

- Suppose the true model contains x_1 and x_2:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$$

- Suppose that, in the estimated model, x_2 is ommited:

$$\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1 + \tilde{u}$$

This leads to a ommited variable bias.

Omitting relevant variables: the simple case

- If x_1 and x_2 are correlated, assume a linear regression relationship between them:

$$x_2 = \delta_0 + \delta_1 x_1 + v$$

- Substituting in the equation of the true model:

$$\begin{aligned} y =& \beta_0 + \beta_1 x_1 + \beta_2 \left(\delta_0 + \delta_1 x_1 + v\right) + u \\ =& \underbrace{\left(\beta_0 + \beta_2 \delta_0\right)}_{\text{Intercept}} + \underbrace{\left(\beta_1 + \beta_2 \delta_1\right)}_{\text{Coefficient of } x_1} x_1 + \underbrace{\left(\beta_2 v + u\right)}_{\text{Error term}} \end{aligned}$$

- If y is only regressed on x_1, and x_1 is correlated with x_2, all coefficients in will be biased.

Omitting relevant variables: the simple case

- Example: Omitting ability in a wage equation

$$wage = \beta_0 + \beta_1 educ + \beta_2 abil + u$$

$$abil = \delta_0 + \delta_1 educ + v$$

Will both be positive

$$wage = (\beta_0 + \beta_2\delta_0) + \boxed{(\beta_1 + \beta_2\delta_1)} educ + (\beta_2 v + u)$$

The return to education β_1 will be <u>overestimated</u> because $\beta_2\delta_1 > 0$. It will look as if people with many years of education earn very high wages, but this is partly due to the fact that people with more education are also more able on average.

Standard assumptions for the multiple regression model (cont.)

- **Assumption MLR.5** (Homoskedasticity): The value of the explanatory variables must contain no information about the variance of the unobserved factors

$$\operatorname{Var}\left(u_i \mid x_{i1}, x_{i2}, \ldots, x_{ik}\right) = \sigma^2$$

- Example: Wage equation

$$\operatorname{Var}\left(u_i \mid \text{ educ }_i, \text{ exper }_i, \text{ tenur }_i\right) = \sigma^2$$

This assumption may also be hard to justify in many cases

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Sampling variances of the OLS slope estimators

- Theorem: Under assumptions MLR.1 – MLR.5:

Variance of the error term

$$Var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1 - R_j^2)}, \qquad j = 1, \ldots, k$$

Total sample variation in explanatory variable x_j:

$$\sum_{i=1}^{n} (x_{ij} - \bar{x}_j)^2$$

R-squared from a regression of explanatory variable x_j on all other independent variables (including a constant)

Components of OLS Variances

1. **The error variance**
   - A high error variance increases the sampling variance because there is more "noise" in the equation.
   - A large error variance doesn't necessarily make estimates imprecise.
   - The error variance does not decrease with sample size.
2. **The total sample variation in the explanatory variable**
   - More sample variation leads to more precise estimates.
   - Total sample variation automatically increases with the sample size.
   - Increasing the sample size is thus a way to get more precise estimates.
3. **Linear relationships among the independent variables**
   - Regress x_j on all other independent variables (including constant).
   - The R-squared of this regression will be higher when x_i can be better explained by the other independent variables.
   - The sampling variance of the slope estimator for x_i will be higher when x_i can be better explained by the other independent variables.
   - Under perfect multicollinearity, the variance of the slope estimator will approach infinity.

The multicollinearity problem

- When two explanatory variables are correlated (colinear), it is difficult to **disentangle** the effect of each variable
- **Example: Debt and Interest Expense**. Consider the following regression model:

$$\text{ROA} = \beta_0 + \beta_1 \times \text{Debt} + \beta_2 \times \text{Interest Expense} + \epsilon$$

- In this case, Debt and Interest Expense are likely to be highly correlated.
- As companies take on more debt, their interest expense generally increases, making it difficult to separate their individual effects on ROA.
- Difficulty in interpreting whether changes in ROA are driven more by Debt or by Interest Expense.
- We might consider removing one of the two variables from the equation (but this may lead to omitted variable bias).

The multicollinearity problem

- What is the effect of multicollinearity on OLS estimates?
- Only the sampling variance of the variables involved in multicollinearity will be inflated; the estimates of other effects may be very precise.
- Note that multicollinearity is not a violation of MLR.3 in the strict sense.
- Multicollinearity may be detected through "**variance inflation factors**"

$$\text{VIF}_j = 1/\left(1 - R_j^2\right)$$

As an (arbitrary) rule of thumb, the variance inflation factor should not be larger than 10

Variances in misspecified models

- The choice of whether to include a particular variable in a regression can be made by analyzing the **tradeoff between bias and variance**.
- Consider the following population and estimated models:
- True population model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$$

- Estimated model 1:

$$\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x_1 + \widehat{\beta}_2 x_2$$

- Estimated model 2:

$$\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1$$

Variances in misspecified models

- Estimated model 1:

$$\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x_1 + \widehat{\beta}_2 x_2$$

$$\text{Var}\left(\widehat{\beta}_1\right) = \sigma^2 / \left[SST_1 \left(1 - R_1^2\right)\right]$$

- Estimated model 2:

$$\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1$$

$$\text{Var}\left(\widetilde{\beta}_1\right) = \sigma^2 / SST_1$$

- Which one is smaller, $\text{Var}\left(\widehat{\beta}_1\right)$ or $\text{Var}\left(\widetilde{\beta}_1\right)$?
  - Conditional on x_1 and x_2, the variance in model 2 is always smaller than that in model 1.
  - However, model 2 will lead to biased estimates (due to omitted variable bias)!
  - We can evaluate the **trade-off between bias and variance**.

Estimating the error variance

- An unbiased estimate of the error variance can be obtained by subtracting the number of estimated regression coefficients from the number of observations.

$$\widehat{\sigma}^2 = \left(\sum_{i=1}^{n} \widehat{u}_i^2 \right) / [n - k - 1]$$

- The number of observations minus the number of estimated parameters is also called the degrees of freedom.
- Theorem: Under MLR. 1 – MLR.5,

$$E\left(\hat{\sigma}^2\right) = \sigma^2$$

Estimation of the sampling variances of the OLS estimators

- The estimated sampling variation (or standard error) of the estimated β_j is given by

$$se\left(\widehat{\beta}_j\right) = \sqrt{\widehat{\mathrm{Var}}\left(\widehat{\beta}_j\right)} = \sqrt{\frac{\widehat{\sigma}^2}{SST_j\left(1 - R_j^2\right)}}$$

- Note that in the above formula we use the unbiased estimate of the error variance, $\widehat{\sigma}^2$.
- Moreover, this formula is only valid under assumptions MLR.1-MLR.5 (in particular, there has to be homoskedasticity).

Efficiency of OLS: The Gauss-Markov Theorem

- Under assumptions MLR. 1 - MLR.5, OLS is unbiased
- However, under these assumptions there may be many other estimators that are unbiased.
- Which one is the unbiased estimator with the smallest variance?
- In order to answer this question one usually limits oneself to linear estimators, i.e. estimators linear in the dependent variable.

$$\tilde{\beta}_j = \sum_{i=1}^{n} w_{ij} y_i \longleftarrow$$

May be an arbitrary function of the sample values of all the explanatory variables; the OLS estimator can be shown to be of this form

Efficiency of OLS: The Gauss-Markov Theorem

- Under assumptions MLR. 1 - MLR.5, the OLS estimators are the best linear unbiased estimators (BLUEs) of the regression coefficients, i.e.

$$\text{Var}\left(\widehat{\beta}_j\right) \leq \text{Var}\left(\widetilde{\beta}_j\right) \quad j = 0, 1, \ldots, k$$

  for all $\tilde{\beta}_j = \sum_{i=1}^{n} w_{ij} y_i$ for which $E\left(\widetilde{\beta}_j\right) = \beta_j, j = 0, \ldots, k$.
- OLS is only the best estimator if MLR. 1 - MLR. 5 hold; if there is heteroskedasticity for example, there are better estimators.

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